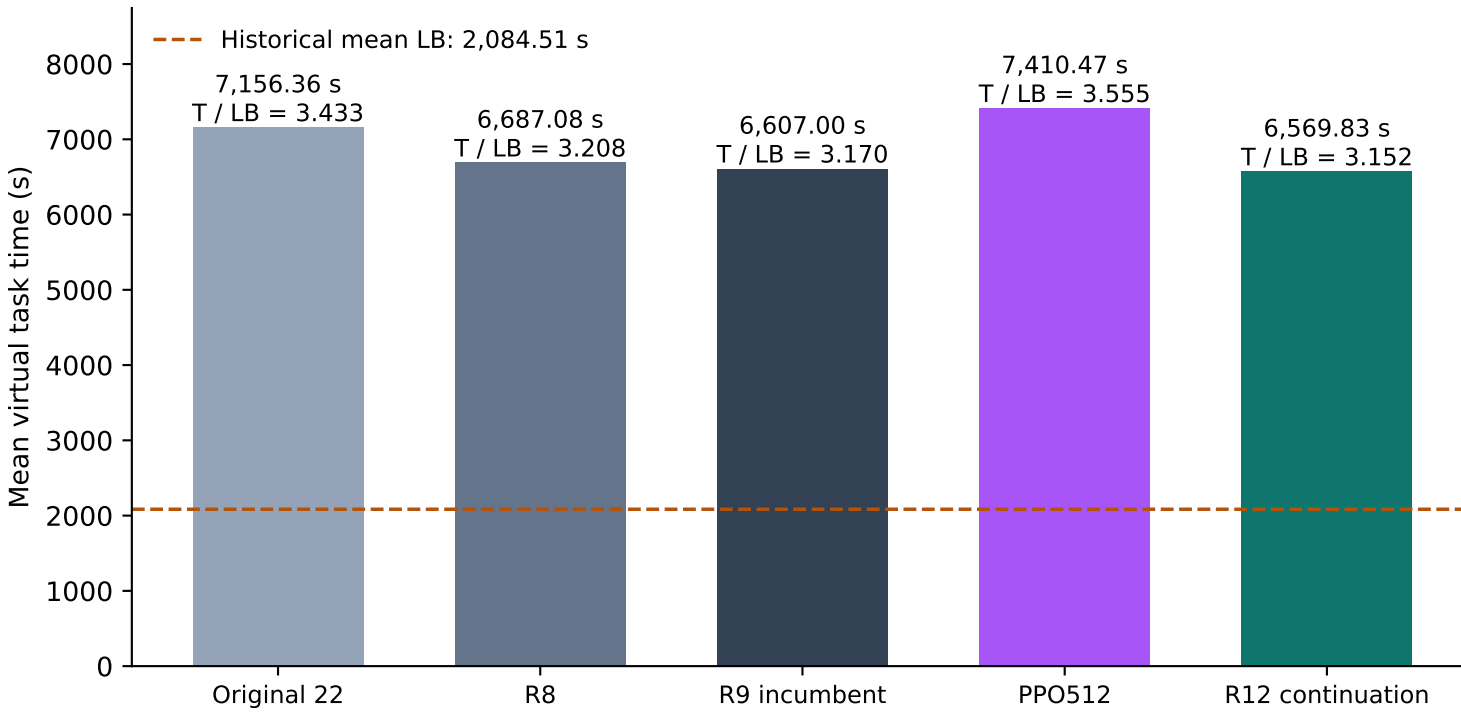
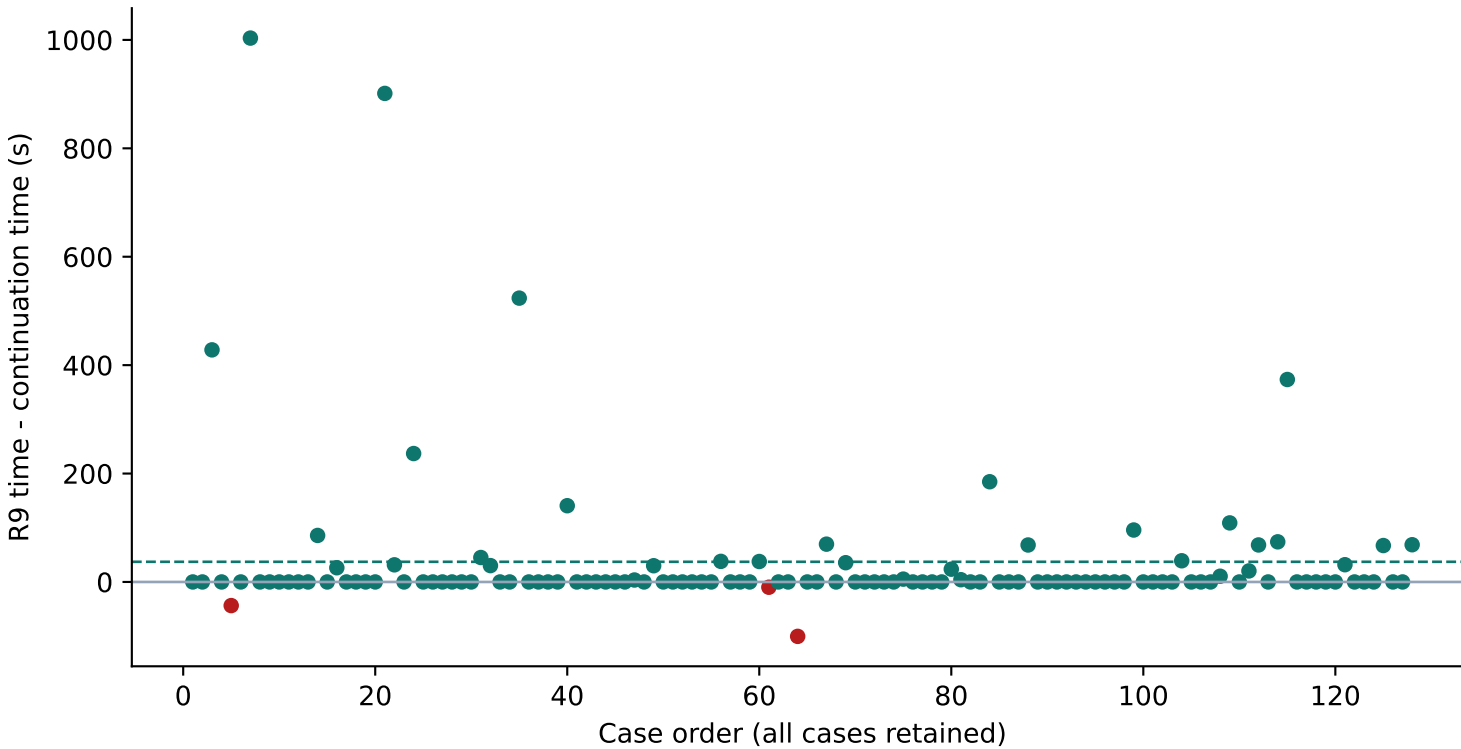


Q4: update certified feasible regions after real silent observations

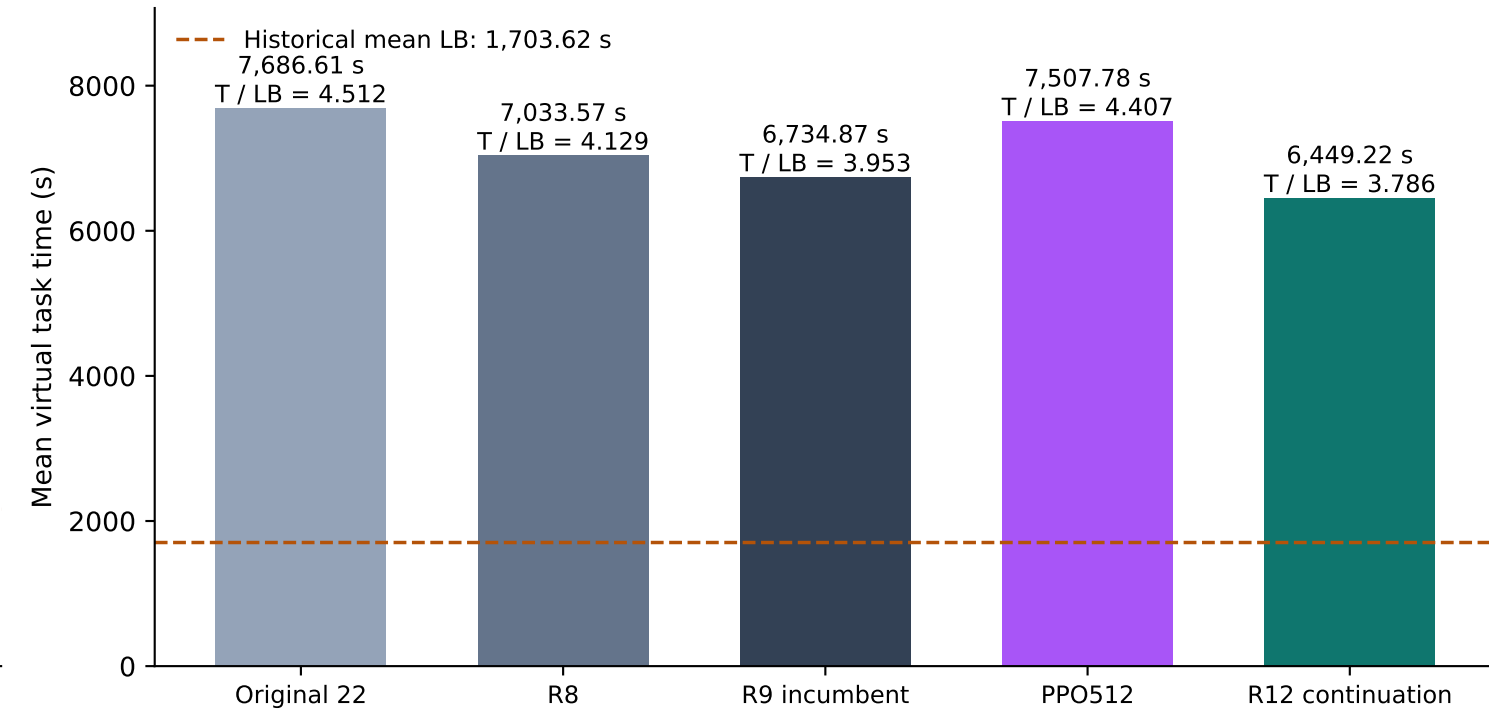
Random: 128 paired cases



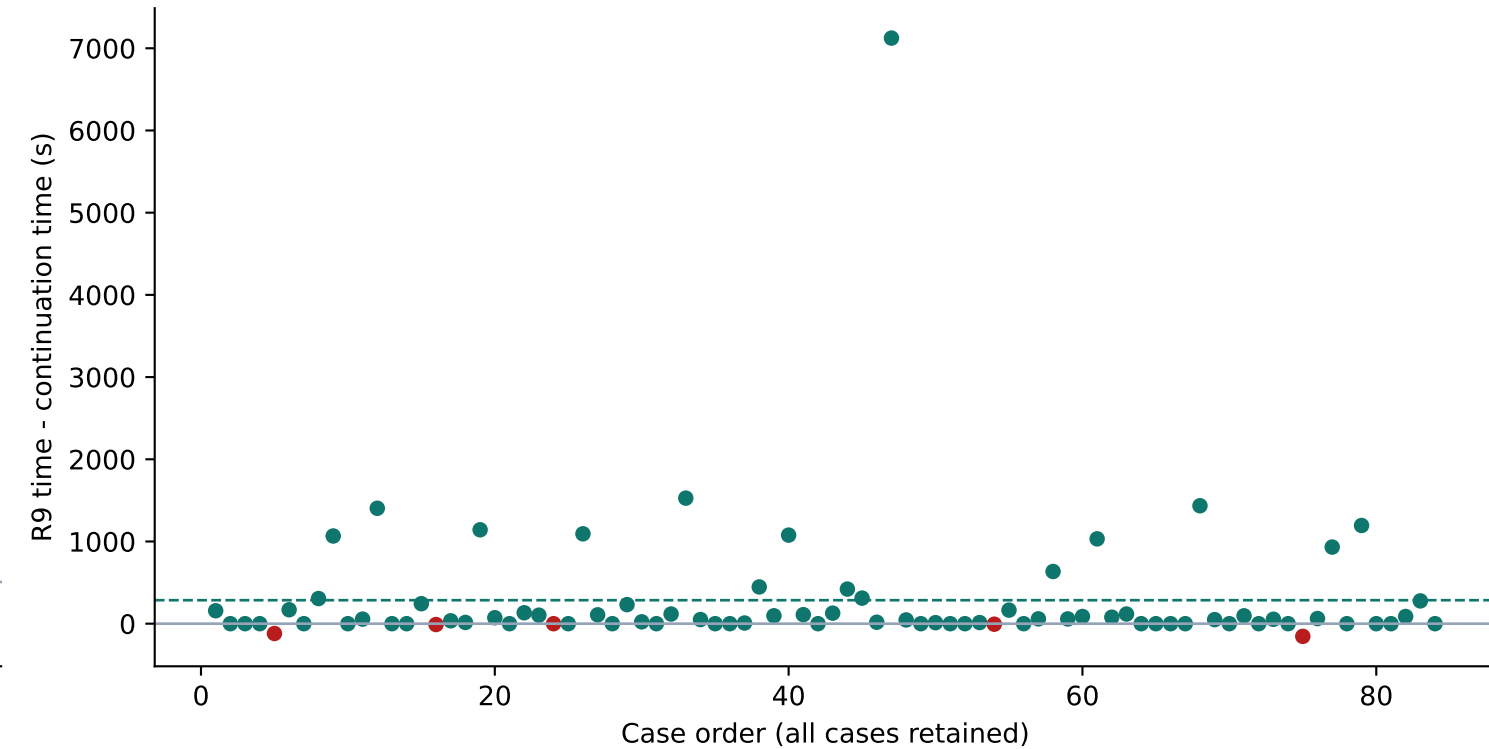
Mean saved 37.17 s (0.563%)
Paired 95% CI: [15.65, 63.30] s



Stress: 84 paired cases



Mean saved 285.65 s (4.241%)
Paired 95% CI: [141.90, 494.03] s



Local generated mixed-source scenarios; all cases cleared. Historical oracle LB remains a relaxation, not an achievable-time claim.